Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/24/2025
PI Name:	Arquilla, Katya Ph.D.		
Project Title:	Investigating Countermeasures for Communication Delays Through the Laboratory-based Exploration Mission Analog (LEMA)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (3) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80309	**Congressional District:**	2
Comments:			
Project Type:		**Solicitation / Funding Source:**	
Start Date:	01/01/2025	**End Date:**	09/30/2025
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per A. Beitman/HFBP (Ed., 8/14/25). NOTE: Start date changed to 01/01/2025 per L. Dang/HFBP. Original start date was 09/25/2024 (Ed., 8/5/25).		
Key Personnel Changes/Previous PI:	None to report.		
COI Name (Institution):	Clark, Torin Ph.D. (University of Colorado, Boulder) Hayman, Allison Ph.D. (University of Colorado, Boulder) Hirshfield, Leanne Ph.D. (University of Colorado, Boulder) Williams, Thomas Ph.D. (Colorado School of Mines)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Communication delays during exploration missions have the potential to reduce the efficiency of collaborative task execution, taxing crewmembers cognitively, physiologically, and psychologically in an environment where maintaining high performance is essential for mission success. Given the relatively short time delay between Mission Control Center (MCC) and the lunar surface (5-14 seconds), both the crew and MCC will be active participants in data gathering, interpretation, and decision-making. These collaborative tasks require the establishment of common understanding in real time between the groups making decisions semiautonomously. In this paradigm, the challenges created by communication delays are centered around unstructured wait time. Once the crew sends a communication, they are left awaiting a response from MCC in a high-stimulation environment; this can lead to additional communications sent by the crew, which can interfere with incoming transmissions and lead to information loss, frustration, and added time to task completion. Given the risks to mission success this creates, it is our goal to develop countermeasures that structure discourse in such a way that both the crew and MCC can continue to augment their understanding of the situation and plan potential actions independently, updating their approach once the next communication is received. We will investigate the efficacy of several countermeasures, including a text-based chatbot that structures the information gathering process, an autonomous system that prompts the crew or MCC to increase their level of understanding of the situation, and audiovisual monitoring that provides additional context for both parties and can be examined during wait time. Each of these countermeasures will first be tested in a Laboratory-based Exploration Mission Analog (LEMA), consisting of multiple spaceflight-relevant task simulations in physical spacecraft mockups, simulated extravehicular activities, collaborative robotic tasks, and tasks in the virtual reality environment. LEMA is restricted to 8-hour maximum simulations and will be used to down-select communication delay countermeasures before the most promising option is deployed in the Human Exploration Research Analog (HERA) to investigate the potential to reduce the challenges presented by communication delays over an extended duration in a relevant isolated, confined environment with a team more reliant on each other and their link with MCC. In both of these settings, we will objectively measure relevant human states – including cognitive load, situation awareness, and stress – using neurophysiological and behavioral measurement techniques to ensure that the structured discourse has the desired effect on human state. Structured discourse enabling the conversion of unproductive wait time into time that can be spent gaining a deeper understanding of the operational environment and developing solutions in real time will improve communication quality, reducing stressors on each human involved in the collaborative task and improving team dynamics within the surface exploration team, MCC, and at the intersection of the two. This mitigates the risk of communication breakdowns and inefficiencies that can lead to dangerous situations in the operational environment. The lunar environment is the first human exploration environment in 50 years to introduce communication delays, but it is not the last. Prolonged wait times will be more pronounced during missions to the Martian surface and beyond, with variable time delays between distributed crews. It is important that we develop guidelines for the most appropriate type of structured communication for a given time delay, and the approaches we are developing here and the methods of evaluation we are implementing will lay the foundation for these guidelines and provide a path forward for future development and validation of communications countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The effectiveness of dynamic, distributed teams operating in high-stress environments is essential for the effective exploration of space, but they are also ubiquitous on Earth across domains. For example, similar teams are deployed to explore the harshest regions of the Earth, search and rescue those in need after natural disasters, and deliver medical care in austere environments, among many other applications. These teams are all challenged by shifting communications infrastructure and would benefit from continued research into how communication challenges affect team performance and cohesion. Additionally, developing technologies to help mitigate the effects of shifting communications infrastructure and support distributed problem-solving could improve performance overall. Across domains, this can maximize efficiency, minimize task overload, and lead to teams that are more equipped to respond to changing conditions.
Task Progress:	Despite the early cancellation of this award (canceled after 9 months of a 5-year award period), we were able to make meaningful progress toward better characterizing the effects of communication delays on team performance. The following sections highlight the primary accomplishments during the 9-month research period. Development of a multi-user task interface: This project included two experimental campaigns executed in two different experimental environments to investigate the same research questions around the effects of communication delays and potential countermeasures. The two test environments were: 1) the Human Exploration Research Analog (HERA) at NASA's Johnson Space Center, which is a collaborative mission analog with multiple PIs; and 2) the Laboratory-based Exploration Mission Analog (LEMA), which is the high-fidelity experimental infrastructure developed at the University of Colorado Boulder specifically for this research effort. Operating in these two environments required the development of a multi-user task that would be implementable and meaningful in both the HERA test environment and in our LEMA experiments. To meet this need, we developed a computer-based task with two distributed teams of two, exposed to asymmetric information, such that they would need to communicate continuously and collaboratively to meet the goals of the task focused on planning the route for an upcoming extravehicular activity (EVA). At the end of this 9-month period of performance, we have completed the task design and pilot-tested it, to assess the level of communication elicited throughout. Experimental design for HERA Campaign 8: This project was originally scheduled to participate in HERA's Campaign 8, which would include four crews of four, each spending 45 days in the analog environment. This effort included completing an Institutional Review Board (IRB)-approved experimental protocol collaboratively with the HERA team, integrating our EVA planning task with planned HERA EVAs. We also selected appropriate hardware for collecting physiological data and compiled survey instruments for assessing crew team dynamics subjectively with guidance from the NASA Behavioral Health and Performance laboratory. Experimental design for LEMA: The LEMA experiments were designed to be tightly controlled to limit confounds in the data but highly realistic and “ecologically valid” to elicit responses from participants analogous to what we might see during a real lunar extravehicular activity (EVA). Experiments were designed for ~8-hour EVAs with teams of two conducting the EVA in simulated spacesuits outside of the laboratory, and a mission control with four participants acting in roles representative of what we would see in the actual mission control during a lunar EVA. At the end of this 9-month period of performance, much of the high-level experimental design was complete, but we were still pilot-testing and iterating on specifics of each task and the detailed experimental procedures.

	Development and pilot-testing of preliminary communications countermeasures: Our team examined different implementations of countermeasures and decided to start by investigating the utility of a chatbot based on a large language model (LLM) that could be queried in real time by the crew during an EVA. The first step toward developing the LLM-based chatbot was to understand the types of inquiries that should be expected during a lunar EVA. We used recordings of communications during Apollo-era lunar EVAs to characterize the types of inquiries we could expect. From these inquiry types, we developed a basic LLM-based chatbot, capable of responding to inquiries during performance of an EVA planning task, and were able to pilot-test this countermeasure. Future work was intended to integrate experimental findings from HERA Campaign 8 into the development of a broader suite of countermeasures, and we still think there is value in deriving requirements for such a countermeasure empirically.
Bibliography Type:	Description: (Last Updated:)